Quantum Mechanics II — Exercise Sheet 4

Fourier observables on the line

Variant: Questions only

Exercise 1 (An exponential packet). On the line let $\psi(x) = a^{-1/2}e^{-|x|/a}$, where $a > 0$. Using the unitary $\hbar$-Fourier convention, find $\widetilde{\psi}(p)$, ΔQ and ΔP. Decide whether this packet saturates the uncertainty bound. You may use $\int_0^\infty e^{-bx}\cos(cx)\,\mathrm{d}x = b/(b^2+c^2)$ for $b > 0$.

Exercise 2 (A bounded momentum window). Let $\widetilde{\psi}(p) = (2b)^{-1/2}\mathbf{1}_{[-b,b]}(p)$ with $b > 0$. (i) Find $\psi(x)$ by Fourier inversion, with its continuous value at zero. (ii) Find its mean free energy and decide whether its position second moment is finite.

Exercise 3 (A momentum shift without changing position). For a normalized $f \in \mathcal{S}(\mathbb{R})$ define $\psi(x) = e^{\mathrm{i}p_0x/\hbar}f(x)$, where $p_0 \in \mathbb{R}$. (i) Find its Fourier transform and relate its position and momentum means and variances to those of f. (ii) Explain why a position density alone cannot determine a momentum distribution.

Exercise 4 (An eigenvalue is not a narrow interval). For the packet in Exercise 2, compute the probability that $|P| \le \varepsilon$ for every $\varepsilon > 0$. (i) Find its limit as $\varepsilon \downarrow 0$. (ii) Compare this with the probability of $P = 0$ and explain why a large probability in a small interval does not produce an eigenvector of P.